

CS1231S Midterms Cheatsheet

by @jmsandiego

1. Proofs

Number System Sets

$\mathbb{N}$: Natural numbers $\{0, 1, 2, \dots\}$ ($\mathbb{Z}_{\geq 0}$)

$\mathbb{Z}$: Integers $\{3, -100, 3^5, \dots\}$

$\mathbb{Q}$: Rational Numbers $\{\frac{1}{2}, -23, 8.6, \frac{-37}{5}, \dots\}$

$\mathbb{R}$: Real Numbers $\{-1, \pi, \sqrt{2}, 4.5, \dots\}$

* 0 is neither negative or positive

* $\mathbb{Z}^{+/-}$ set of all positive or negative integers

Terminologies

Axiom/Postulate: A statement assumed to be true w/o proof. Basic building blocks of theorems

Theorem: A mathematical statement proved using rigorous mathematical reasoning. Major / important result.

Lemma: A small theorem minor result to help prove a theorem.

Corollary: Simple deduction from Theorem.

Conjecture: A statement believed to be true, but no proof yet.

Basic Properties of Integers

Closure: Closed under $+/ \times$

$x + y \in \mathbb{Z}$ and $xy \in \mathbb{Z}$

Commutativity: Commutative under $+/ \times$

$x + y = y + x$ and $xy = yx$

Associativity: Associative under $+/ \times$

$x + y + z = (x + y) + z = x + (y + z)$ and $xyz = (xy)z = x(yz)$

Distributivity:

$x(y + z) = xy + xz$

Trichotomy: Exactly one of the following is true

$x = y$ or $x < y$ or $x > y$

Definitions of Number

Even and Odd Integers

Let $n \in \mathbb{Z}$

n is even $\leftrightarrow \exists k \in \mathbb{Z} \ n = 2k$

n is odd $\leftrightarrow \exists k \in \mathbb{Z} \ n = 2k + 1$

Divisibility

Let $d, n \in \mathbb{Z} \wedge d \neq 0$

$d|n \leftrightarrow \exists k \in \mathbb{Z} \ n = dk$

* d divides n

Prime / Composite

- **Byte:** 8-bits
- **Nibble:** 4-bits
- **Word:** 1 / 2 / 4 bits etc. based on architecture

$N_{bits} = 2^N \text{ values}$

$M_{values} = \log_2 M$ *take ceiling value

Decimal (base 10)

Symbols = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Binary (base 2)

Symbols = 0, 1

* represented by prefix 0b

Octal (base 8)

Symbols = 0, 1, 2, 3, 4, 5, 6, 7

* represented by prefix 0

Hexadecimal (base 16)

Symbols = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

* represented by prefix 0x

Conversions

- Base-R to Decimal
 - multiply each digit with respective positional weight. $(\text{digit} \times R^{\text{position}}) + \dots$
- Decimal to Base-R
 - **whole number:** Repeated division by R
* to find remainder multiply fractional value with the divisor / R
 - **fractions:** Repeated multiplication by R Method.
* Until fractional product is 0 or desired number of decimal decimal places
- Between Bases
 - partition in groups of $\text{curr}R^N = \text{convert}R$

ASCII

American Standard Code for Information Interchange

7 bits plus 1 bit for odd / even parity (for error checking)

Integers (0 - 127) are interchangeable with Characters

Negative / Signed Numbers

- **Sign-and-Magnitude**
 - Sign is represented by the MSB '0' is positive
 - Range for 8-bit:
 $-2^{n-1} - 1 = -127_{10}$ to $2^{n-1} - 1 = +127_{10}$
 - Two zeroes: $00000000 = +0_{10}$ & $10000000 = -0_{10}$
- **1s Complement**
 - Negated value of x: $-x = 2^n - x - 1$ / Flip the bits
 - Range for 8-bit:
 $-2^{n-1} - 1 = -127_{10}$ to $2^{n-1} - 1 = +127_{10}$
 - Two zeroes: $00000000 = +0_{10}$ & $11111111 = -0_{10}$

- **2s Complement**

- Negated value of x: $-x = 2^n - x$ / 1s complement + 1 / Copy from LSB until first '1' then invert rest till MSB
- Range for 8-bit:
 $-2^{n-1} = -128_{10}$ to $2^{n-1} - 1 = +127_{10}$
- One zero: $00000000 = +0_{10}$

- **Excess Representation**

- Range of values distributed evenly between positive and negative values
- Excess-M starts from $00..00 = -M$ onwards
- To evenly distribute, use the -2^{n-1} range as M value

Addition / Subtraction 1s & 2s Complement

- **1s Complement**

1. Perform binary Addition
2. Carry out add 1 to result
3. Check overflow if MSB result is opposite sign of A & B

- **2s Complement**

1. Perform binary Addition
2. Ignore carry out of MSB
3. Check overflow 'carry in/out' of MSB different / like in 1s complement

Fixed-Point Representation

Number of bits allocated for whole and fractional part are fixed For example: If 2s complement is used:

$011010.112_s = 26.75_{10}$

$111110.112_s = -000001.012 = -1.25_{10}$

Limitations: limited range and precision and immutable once implemented

Floating-Point Representation (IEEE 754)

Allows to represent very large or small numbers compared to fixed point such as:

0.23×10^{23} (very large positive number)

0.5×10^{-37} (very small positive number)

-0.2397×10^{-18} (very small negative number)

IEE 754 Single precision (32-bits)

Consists of 3 main components:

- Sign (1-bit)
 - 0/1: Positive / Negative
- Exponent (8-bit w/ excess-127)
 - $exponent + 127 = 129 = 10000001_2$
- Mantissa (fraction) (23-bit)
 - normalised with an implicit leading bit 1
 - $110.1_2 \rightarrow 1.101_2 \times 2^2$ (101 is stored in mantissa)

11. Processor Datapath

MIPS Instruction Execution

Fetch Stage

1. Fetch the instruction from memory with **Program Counter (PC)**
** PC is a special register*
2. Increment PC by 4 to get next address
3. Output is provided to Decode / Operand Fetch Stage using the Instruction Memory. ** address $\rightarrow$ binary instruction*

PC is read during first half of clock period and updated at next rising clock edge

Decode / Operand Fetch Stage

Gather data from instruction field:

1. Input is the binary instruction from Fetch Stage
2. Read opcode to determine instruction format and field lengths
3. Fetch data from necessary registers into the Write Register according to RegDst Signal (Rt / Rd)
4. Output is the necessary operands to be conducted in ALU Stage

Multiplexer

Selects a specific input from multiple input lines. Consists of:

- Inputs: n lines of same width (32-bits)
- Control: m bits where $n = 2^m$. Specifies which input to choose
- Output: The selected input based on i^{th} input is $control = i$

ALU Stage / Execution Stage

Performs the following operations:

- Arithmetic, Shifting & Logical
 - Memory operation - address calculation
 - Branch operation - register comparison & target address calculation
1. Input is the two 32-bits numbers from Decode Stage
2nd Operand is chosen according to ALUSrc signal
 2. With the help of 4-bit ALUControl signal, the respective operation is conducted
 3. Outputs ALUresult & a 1-bit signal isZero

Branching in ALU

1. Identify the branch outcome through isZero to handle equal / not equal
2. Calculate Branch address with PC and Offset (Immediate value)
3. Output will be used by PCSrc signal

Memory Stage

Mainly for memory read and write instructions **lw** / **sw**.

Other instructions remain idle in this stage and instead ALU result passed into Register Write Stage

1. Input computation result to be used as Memory Address. Also Write Data for sw (Register Data 2)
2. Read or write data on the memory address depending on MemRead/Write Signal
3. Output memory address content to Register Write stage (lw)

Register Write Stage

Routes the correct result into the Write Data in register file.

Instructions like stores, branches and jumps remain idle in this stage.

1. Input computation result from Memory or ALU, which is specified by MemToReg signal
2. Outputs the selected result into the Write Register in the register file

12. Processor Control

Control Signals

- **RegDst** - (@Decode/OperandFetch) Selects the destination of WriteRegister in register file
 - 0/1: WriteRegister = Ins[20:16] (rt) / Inst[15:11] (rd / immediate)
- **RegWrite** - (@Decode/OperandFetch) Indicates if writing on WriteRegister is enabled
 - 0/1: No register write / WD write to WR
- **ALUSrc** - (@ALU) Determines the 2nd ALU input
 - 0/1: Read Data 2 / SignExtend(Inst15:0) (Sign extended Immediate)
- **ALUcontrol** - (@ALU) Determines the type of operation to execute
 - 0000: AND
 - 0001: OR
 - 0010: add
 - 0110: subtract
 - 0111: slt
 - 1100: NOR

- **MemRead** - (@Memory) Indicates if read from memory using Address returned by ALU
 - 0/1: No read / Reads memory with Address returned by ALU
- **MemWrite** - (@Memory) Indicates if write to memory using RD2 of Register File
 - 0/1: No memory write / Writes RD2 into the memory in Address returned by ALU
- **MemToReg** - (@RegWrite) Indicates which one to write back into WriteData in register file
 - 0/1: WriteData = ALU result / Memory read data
- **PCSrc** - (@ALU) Indicates whether to take branch address or not
 - 0/1: $PC = PC + 4$ / $PC = \text{SignExt}(\text{Inst}[15 : 0]) \ll 2 + (PC + 4)$
 - PCSrc = isBranch AND isZero

ALU

** Note: operation is determined by the last 2-bits **
Subtraction has Cin = 1 for bit-0

Multilevel Decoding

Used to generate the ALUcontrol signal. *(Simpler and more efficient design than Brute Force approach)*

1. Generate a 2-bit ALUop signal from opcode
2. Use ALUop and Function code (for R-type) to generate 4-bit ALUcontrol signal

Control Combinational Circuit

Instruction Execution

How the implementation of fetch, decode, memory, write, etc. are actually done

Mainly three things:

1. Read contents from register / memory
2. Perform some form of computation
3. Write results to register / memory

Single Cycle Execution

All performed within a clock period.

disadvantage: all instructions will take the same time as the slowest one.

Multi Cycle Execution

Break up instructions into execution steps (Fetch, Decode, ALU Operation, Mem read/write, Register Write). Each execution is one clock cycle. **disadvantage:** time taken depends on instructions and may not necessarily be faster.

Datapath & Control Diagrams